

Date: 2026-05-21
Request ID: 100048574

Microwave				
Element(s)	Ca			
SOP#				
Acid Cocktail	6 mL HCl, 2 mL HNO3			
Rack	7			
Vessel	PTFE			
Stir	0%			
Time (min)	Power (W)	T1 (°C)	T2 (°C)	P (bar)
15	1200	180	65	110
15	1500	240	65	110
30	1500	240	65	110
sample(s)	weight (g)	volume (mL)		
200128071_1	0.1097	250		
200128071_2	0.1080	250		
200128072_1	0.1096	250		
200128072_2	0.1031	250		
200128073_1	0.1056	250		
200128073_2	0.1041	250		

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200128071_1	50/4	1.735	49424.57	4.94
200128071_2	50/4	1.737	50260.42	5.03

ca result: 49842.49 ppm

ca result: 4.98 %

ca oxide factor: 1.399

ca oxide result: 6.97 %

ca lot: *Manufacturer: Assurance, 28-35CAY exp: 2026-10-30*

Note(s):

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200128072_1	50/2	1.589	90613.59	9.06
200128072_2	50/2	1.521	92204.17	9.22

ca result: 91408.88 ppm

ca result: 9.14 %

ca oxide factor: 1.399

ca oxide result: 12.79 %

ca lot: *Manufacturer: Assurance, 28-35CAY exp: 2026-10-30*

Note(s):

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200128073_1	50/2	2.502	148082.39	14.81
200128073_2	50/2	2.311	138748.80	13.87

ca result: #####

ca result: 14.34 %

ca oxide factor: 1.399

ca oxide result: 20.06 %

ca lot: *Manufacturer: Assurance, 28-35CAY exp: 2026-10-30*

Note(s):

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A] - [C-A]) / (B-A) * 100\%$$

$$ppm\ M+ = [conc.][volume][dilution] / [weight]$$

$$\%M+ = [conc.][volume][dilution] / ([weight] * 10,000)$$

$$\%MO = [oxide\ factor] * \%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1 - (\%LOI) / 100])$$

$$\%M+ \ [dried] = \%M+ / ([1 - (\%LOI) / 100])$$

$$\%MO \ [dried] = \%MO / ([1 - (\%LOI) / 100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS]) / ([weight])$$

$$\%Cr_2O_3 = 1.462 * [\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$